

Project Title: MindMerge SmartCampus – School Management System

Project Objectives

Develop a centralized platform for:

- ✓ Student Management
- ✓ Teacher Management
- ✓ Attendance Tracking
- ✓ Timetable Management
- ✓ Exams & Results
- ✓ Parent Login Portal
- ✓ Notifications
- ✓ Fees Management
- ✓ Reports & Analytics
- ✓ Admin Dashboard

Tech Stack (Detailed)

Frontend

HTML5

Used for:

- Structure of webpages
- Forms
- Tables
- Dashboards

Examples:

Student Registration Form

Attendance Form

Login Pages

CSS3

Used for:

- Styling
- Responsive UI
- Layouts

Features:

Animations

Cards

Dashboards

Responsive Design

Bootstrap 5

Used for:

Navbar

Cards

Tables

Forms

Dashboard UI

Responsive design

Example:

Bootstrap Grid System

Modal

Alerts

Pagination

JavaScript

Used for:

Form Validation

Dynamic Data Loading

Notifications

AJAX Requests

Attendance Updates

Backend

PHP

Handles:

Authentication

Database Operations

CRUD Functions

Reports

Session Handling

Examples:

Login

Student Registration

Attendance Save

Generate Results

Database

MySQL

Stores:

Students

Teachers

Attendance

Fees

Subjects

Timetables

Results

Parents

Notifications

Additional Tools

XAMPP

Provides:

Apache Server

PHP Runtime

MySQL Server

Git + GitHub

Version control

VS Code

Development environment

 Major Modules

Module 1: Authentication & Authorization

Purpose:

Manage user access

Roles:

Admin

Teacher

Student

Parent

Features:

- ✓ Login
 - ✓ Logout
 - ✓ Forgot Password
 - ✓ Role Management
 - ✓ Session Handling
-

Files:

login.php

logout.php

forgot-password.php

auth.php

session.php

Module 2: Dashboard Module

Shows overview

Features:

Total Students

Teachers Count

Attendance

Recent Notifications

Results

Files:

dashboard.php

stats.php

widgets.php

Module 3: Student Management

Features:

- ✓ Add Student
 - ✓ Update Student
 - ✓ Delete Student
 - ✓ Search Student
 - ✓ Student Profiles
-

Database:

students

Files:

add-student.php

edit-student.php

student-list.php

student-profile.php

Module 4: Teacher Management

Features:

Add Teacher

Assign Subjects

Manage Salary

Update Details

Files:

add-teacher.php

teacher-list.php

teacher-profile.php

Module 5: Attendance Management

Features:

Daily Attendance

Reports

Monthly Reports

Export Attendance

Files:

mark-attendance.php

attendance-report.php

Module 6: Timetable Management

Features:

Class Schedule

Teacher Schedule

Exam Schedule

Files:

timetable.php

manage-schedule.php

Module 7: Exam & Results Management

Features:

Create Exams

Marks Entry

Result Generation

Rank Calculation

Files:

add-marks.php

results.php

generate-result.php

Module 8: Parent Portal

Features:

Parents can view:

Attendance

Results

Fees

Notifications

Files:

parent-login.php

parent-dashboard.php

Module 9: Notifications Module

Features:

Send Notices

Alerts

Event Notifications

Files:

notifications.php

send-alert.php

Module 10: Fees Management

Features:

Fee Collection

Pending Fees

Receipts

Files:

fees.php

payment-history.php

Module 11: Reports Module

Generate:

Attendance Reports

Student Reports

Fees Reports

Exam Reports

Files:

reports.php

export-pdf.php

Module 12: Admin Panel

Controls:

Users

Roles

Settings

Permissions

Files:

admin.php

settings.php

manage-users.php

Complete Folder Structure

MindMergeSmartCampus/

|

|— assets/

| |— css/

| | style.css

| | dashboard.css

| |

| |— js/

| | app.js

| | validation.js

| |

| |— images/

|

|— config/

| database.php

| config.php

|

|— auth/

| login.php

| logout.php

| forgot-password.php

| session.php

|

|— dashboard/

| dashboard.php

| stats.php

|

|— students/

| add-student.php

| edit-student.php

| delete-student.php

| student-list.php

| student-profile.php

|

|— teachers/

| add-teacher.php

| teacher-list.php

| teacher-profile.php

|

|— attendance/

| mark-attendance.php

| attendance-report.php

|

|— timetable/

| timetable.php

| manage-schedule.php

|

|— exams/

| add-marks.php

| generate-results.php

| exam-schedule.php

|

|— parents/

| parent-login.php

| parent-dashboard.php

|

|— fees/

| fees.php

| payment-history.php

|

|— notifications/

| notifications.php

| send-alert.php

|

|— reports/

| reports.php

| export-report.php

|

|— admin/

| admin.php

| settings.php

| manage-users.php

|

|— api/

| student-api.php

| attendance-api.php

|

|— uploads/

| student-images/

| documents/

|

|— index.php

└─ README.md

Database Tables

users

students

teachers

attendance

subjects

classes

fees

payments

results

exams

notifications

parents

timetable

roles

permissions

 Module Linkups / Workflow

LOGIN



Dashboard



| Student Module

|

| Teacher Module |

| Attendance Module |

| Timetable Module |

| Exam Module |

| Fees Module |

| Notifications |

↓

Reports

↓

Admin Control

Detailed Connections:

Student Module

↓

Attendance Module

↓

Exam Module



Results Module



Parent Portal



Reports

Teacher Module



Subject Assignment



Attendance



Timetable



Exam Management

Fees Module



Payment Records



Receipts



Reports

Admin



Controls Everything

 Project Development Flow (Recommended)

Week 1

Authentication + Database + Dashboard

Week 2

Student + Teacher Modules

Week 3

Attendance System

Week 4

Timetable + Subjects

Week 5

Exam & Results

Week 6

Fees + Parent Portal

Week 7

Reports + Notifications

Week 8

Testing + Deployment + Documentation

Users of System

The system contains 4 major users

1. Admin

Admin controls everything.

Admin can:

Add students

Add teachers

Manage classes

Manage subjects

Manage attendance

Manage exams
Manage fees
Generate reports
Control users

Admin has full access.

2. Teacher

Teacher can:

Mark attendance
Upload marks
View timetable
Manage students
Send notifications

3. Student

Student can:

View attendance
View results
Check timetable
View notices

4. Parent

Parents can:

Monitor attendance
Check marks
Track fees
Receive notices



Complete Project Workflow

The system works in this order:

Login



Dashboard



Student Registration



Teacher Assignment



Attendance Management



Timetable Management



Exam Management



Result Generation



Fees Management



Notifications



Reports



Parent Monitoring



Detailed Module Explanation



Module 1: Authentication System

Purpose:

Provides secure login.

Users:

Admin

Teacher

Student
Parent

Features:

- Login
- Logout
- Password reset
- Session handling

Example:

Admin enters:

Username
Password

System verifies database and gives dashboard access.

Module 2: Dashboard Module

After login users see:

Example dashboard:

Total Students: 1200
Teachers: 70
Today's Attendance: 95%
Upcoming Exams: 5
Pending Fees: 50

Purpose:

Provide quick overview.

Module 3: Student Management

Stores student information.

Example:

Student Details:

Student ID

Name

Class

Section

DOB

Address

Parent Contact

Photo

Operations:

Create

Read

Update

Delete

(CRUD Operations)

Module 4: Teacher Management

Stores:

Teacher Name

Qualification

Assigned Subjects

Salary

Contact Details

Admin assigns teachers to classes.

Example:

Teacher:

Rahul

Subject:

Math

Class:
10th A

Module 5: Attendance Management

Teachers mark attendance.

Example:

Student	Status
Ravi	Present
Sai	Absent

Attendance stored in database.

Monthly reports generated automatically.

Module 6: Timetable Management

Creates schedules.

Example:

Period	Subject
1	Maths
2	English
3	Science

Visible to:

Students

Teachers

Parents

Module 7: Exam Management

Admin creates exams:

Example:

Mid Exam

Final Exam

Unit Test

Teachers upload marks.

Module 8: Result Generation

System calculates:

Total Marks

Percentage

Grade

Rank

Example:

Total:

480/500

Percentage:

96%

Grade:

A+

Results generated automatically.

Module 9: Fees Management

Tracks:

Paid Fees
Pending Fees
Receipts
Due Dates

Parents can monitor payments.

Module 10: Notification Module

Sends alerts.

Example:

Exam tomorrow
Holiday notice
Fee reminder
Parent meeting

Module 11: Parent Portal

Parents log in and monitor:

Attendance
Marks
Fees
Notifications

This improves communication between school and parents.

Module 12: Reports Module

Generate reports like:

Attendance Report

Exam Report

Student Report

Fee Report

Teacher Report

Export:

PDF

Excel

CSV

Database Flow

Example database relationship:

Students



Attendance



Exams



Results



Parents

Teachers



Subjects



Timetable



Attendance

Fees



Payments



Reports



Real-Time Module Connections

Example:

When a teacher marks attendance:

Teacher Login



Attendance Module



Database Updated



Parent Dashboard Updated



Reports Updated

When marks entered:

Teacher



Marks Entry



Result Calculation



Student Dashboard



Parent Dashboard

Everything is interconnected.



Example Real-Life Scenario

Suppose:

Student: Arjun

Morning:

Teacher marks attendance

↓

Attendance saved

↓

Parent sees attendance

↓

Exam marks uploaded

↓

Results generated

↓

Parent checks report

All activities happen through one system.

Additional Features Explanation

1. 🌙 Dark Mode

Purpose:

Allow users to switch between **Light Mode** and **Dark Mode** for better accessibility and reduced eye strain.

Users:

- Admin
- Teacher
- Student
- Parent

Features:

- ✓ Theme switch button
- ✓ Save user preference
- ✓ Automatic dark mode

Tech Required:

Frontend:

HTML

CSS

Bootstrap

JavaScript

Database:

users table

theme_preference column

Files:

assets/css/light.css

assets/css/dark.css

assets/js/theme.js

settings/theme.php

Workflow:

User Login



Select Theme



Save Preference



Load Selected Theme

2. QR Attendance System

Purpose:

Students scan QR code to mark attendance.

Features:

- ✓ QR generation
- ✓ QR scanning
- ✓ Auto attendance marking

Tech Stack:

PHP

JavaScript

QR Library

MySQL

Libraries:

phpqrcode

html5-qrcode

Files:

attendance/

generate-qr.php

scan-qr.php

qr-attendance.php

Database:

qr_logs
attendance

Workflow:

Generate QR



Student Scan



Attendance Saved



Reports Updated

3. 😊 Face Recognition Attendance

Purpose:

Attendance via face detection.

Features:

- ✓ Capture image
- ✓ Match face
- ✓ Mark attendance

Tech Stack:

Frontend:

JavaScript
Camera API

AI:

Python
OpenCV
Face Recognition

Backend:

PHP API
MySQL

Files:

face-recognition/
capture.py
train_model.py
recognize.py

Database:

face_data
attendance

Workflow:

Camera
↓
Face Detection
↓
Recognition
↓
Attendance

4. AI Student Performance Prediction

Purpose:

Predict future student performance using attendance, marks, and activities.

Example:

Attendance: 60%
Marks: Low

Prediction:
Risk of failure

Tech:

AI:

Python
Machine Learning
Scikit-Learn
Pandas
NumPy

Model:

Linear Regression
Random Forest

Files:

ai/
 train_model.py
 prediction.py

Database:

student_performance
predictions

Workflow:

Attendance
Marks
Activities



ML Model



Prediction

5. ➡️ SMS Notifications

Purpose:

Send:

- ✓ Fee reminders
- ✓ Attendance alerts
- ✓ Exam notifications

APIs:

Examples:

Twilio
Fast2SMS
MSG91

Files:

notifications/
send_sms.php

Workflow:

Event
↓
SMS API
↓
Parent Receives SMS

6. Email Notifications

Purpose:

Send:

Results
Fee reminders
Events
Announcements

Tools:

PHPMailer

SMTP

Gmail API

Files:

notifications/

send_email.php

7. Mobile Responsive Design

Purpose:

System works on:

✓ Mobile

✓ Tablet

✓ Desktop

Tech:

Bootstrap

CSS Media Queries

Implementation:

Responsive Navbar

Responsive Tables

Responsive Forms

8. Multi-language Support

Purpose:

Support:

English

Telugu

Hindi

Tamil

Implementation:

Language files:

lang/

english.php

telugu.php

hindi.php

Database:

user_language

Workflow:

User selects language



System loads translations

9. Export PDF Reports

Purpose:

Generate:

✓ Student reports

✓ Attendance reports

✓ Fee reports

Libraries:

TCPDF

FPDF
DomPDF

Files:

reports/
export-pdf.php

Output:

PDF Reports
Certificates
Result Sheets

10. Parent & Teacher Chat System

Purpose:

Real-time communication.

Features:

- ✓ Messaging
- ✓ Notifications
- ✓ File sharing

Tech:

PHP
AJAX
WebSocket
MySQL

Database:

messages
chat_history

Files:

chat/
 send.php
 receive.php

Workflow:

Parent

↔

Teacher

11. 🚌 Live Bus Tracking

Purpose:

Track school buses.

Features:

- ✓ Real-time location
- ✓ Estimated arrival
- ✓ Parent notifications

Tech:

Google Maps API

GPS

PHP

Files:

transport/
 bus_tracking.php

Database:

bus_locations

routes

Workflow:

GPS Device
↓
Location Update
↓
Map Display

12. AI Chatbot Support

Purpose:

Provide instant help.

Examples:

Students ask:

When is exam?

What is my attendance?

Show timetable

Chatbot responds automatically.

Tech:

Python

NLP

OpenAI API

Dialogflow

PHP

Files:

chatbot/

chatbot.php

api.py

Workflow:

User Question



AI Model



Response



Additional Folder Structure

MindMergeSmartCampus/

- ai/
 - prediction.py
 - train_model.py
- chatbot/
 - chatbot.php
 - api.py
- chat/
 - send.php
 - receive.php
- notifications/
 - send_sms.php
 - send_email.php
- transport/
 - bus_tracking.php
- face-recognition/
 - recognize.py
 - capture.py
- reports/
 - export-pdf.php
- lang/
 - english.php
 - telugu.php

| hindi.php
|
|— attendance/
| generate-qr.php
| scan-qr.php

Complete Feature Linkups

Student



Attendance



AI Prediction



Reports



Parent Notification

Teacher



Attendance



Marks



Results



SMS + Email

Bus GPS



Live Tracking



Parent Dashboard

Chat



Teacher ↔ Parent

AI Chatbot



Answer Questions

Team Structure



Team Lead (Member 1)

Role:

Manage complete project architecture and integration

Responsibilities:

- ✓ Requirement analysis
- ✓ Database design
- ✓ Module integration
- ✓ GitHub management
- ✓ Deployment
- ✓ Team coordination

Work:

Project Setup
Folder Structure
Database Architecture
API Connections
Final Integration
Deployment
Documentation

Tech:

PHP
MySQL

Git
GitHub
XAMPP

Deliverables:

ER Diagram
Database
Architecture Design
Deployment



Member 2 — Authentication + Dashboard + Dark Mode

Modules:

Login
Logout
Session
Role Management
Dashboard
Dark Mode

Tasks:

Week 1:

Login UI
Authentication
Sessions

Week 2:

Dashboard
Dark Mode

Files:

auth/
dashboard/



Member 3 — Student Management + Parent Portal

Modules:

Student CRUD
Student Profiles
Parent Login
Parent Dashboard

Tasks:

Build:

- ✓ Add student
- ✓ Edit student
- ✓ Parent access

Files:

students/
parents/

Database:

students
parents



Member 4 — Teacher Management + Timetable

Modules:

Teacher CRUD
Subjects
Timetable
Class Allocation

Files:

teachers/
timetable/

Tables:

teachers
subjects
timetable



Member 5 — Attendance System + QR Attendance

Modules:

Attendance
QR Generation
QR Scanning
Reports

Tasks:

Develop:

- ✓ Manual attendance
- ✓ QR attendance

Files:

attendance/
generate-qr.php
scan-qr.php

Libraries:

phpqrcode
html5-qrcode



Member 6 — Exam + Results + AI Prediction

Modules:

Exams
Marks Entry
Results
AI Prediction

Tasks:

Build:

- ✓ Marks upload
- ✓ Result generation
- ✓ Student performance prediction

Tech:

PHP
Python
Scikit-learn

Files:

exams/
ai/



Member 7 — Fees + Reports + PDF Export

Modules:

Fees
Payments
Receipts
Reports
PDF Export

Tasks:

Generate:

- ✓ Fee reports
- ✓ Attendance reports
- ✓ PDF reports

Libraries:

DOMPDF
FPDF

Files:

fees/
reports/



Member 8 — Notifications (SMS + Email)

Modules:

SMS
Email

Alerts
Announcements

Integrations:

Twilio
MSG91
PHPMailer
SMTP

Files:

notifications/

Tasks:

Send:

- ✓ Exam alerts
- ✓ Fee reminders
- ✓ Attendance alerts



Member 9 — Chat System + AI Chatbot

Modules:

Parent Teacher Chat
AI Chatbot

Tasks:

Build:

- ✓ Real-time messaging
- ✓ AI chatbot

Tech:

PHP
AJAX
WebSocket
Python
OpenAI API

Files:

chat/
chatbot/



Member 10 — Face Recognition + Bus Tracking + Responsive Design

Modules:

Face Recognition
Live Bus Tracking
Responsive Design

Tasks:

Build:

- ✓ Face attendance
- ✓ GPS tracking
- ✓ Mobile responsiveness

Tech:

OpenCV
Python
Google Maps API
Bootstrap

Files:

face-recognition/



8 Weeks Development Plan

Week	Tasks
Week 1	Requirement Analysis + DB + Authentication
Week 2	Dashboard + Students + Teachers
Week 3	Attendance + Timetable
Week 4	Exams + Results + Fees
Week 5	Parent Portal + Reports + Notifications
Week 6	QR + AI Prediction + Chat
Week 7	Face Recognition + Bus Tracking + Chatbot
Week 8	Testing + Integration + Deployment

Module Dependencies (Who works with whom)

Member 2 (Auth)



All Members

Member 3 (Students)



Member 5 (Attendance)



Member 6 (Results)



Member 7 (Reports)



Member 8 (Notifications)

Member 4 (Teachers)



Member 5 (Attendance)



Member 6 (Exams)

Member 9 (Chat)



Member 3 (Parents)



Member 4 (Teachers)

Member 10 (Face Recognition)



Member 5 (Attendance)



Suggested Team Ownership Structure

Member1 → Project Lead + Database + Deployment

Member2 → Auth + Dashboard + Dark Mode

Member3 → Student + Parent

Member4 → Teacher + Timetable

Member5 → Attendance + QR

Member6 → Exam + AI Prediction

Member7 → Fees + Reports + PDF

Member8 → SMS + Email Notifications

Member9 → Chat + AI Chatbot

Member10 → Face Recognition + Bus Tracking + UI